ASSIGNMENT 2

SECI1143: PROBABILITY & STATISTICAL DATA ANALYSIS

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SECTION: 02

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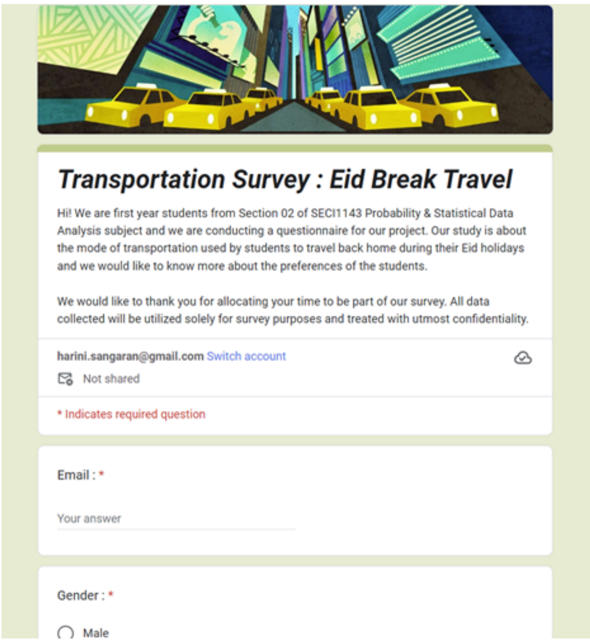
INTRODUCTION

We conducted a study on the transportation used by students who are pursuing their higher studies to return to their hometown during the past Eid break. Our questionnaire was not limited only to UTM students but also was open to students from all over Malaysia who are studying in either government or private institutes. We used questionnaire methodology to conduct our survey and our questionnaire was created using Google Form. The questions asked in our questionnaire are as images attached below.



Transportation Survey : Eid Break Travel

Hi! We are first year students from Section 02 of SECI1143 Probability & Statistical Data Analysis subject and we are conducting a questionnaire for our project. Our study is about the mode of transportation used by students to travel back home during their Eid holidays and we would like to know more about the preferences of the students.

We would like to thank you for allocating your time to be part of our survey. All data collected will be utilized solely for survey purposes and treated with utmost confidentiality.

harini.sangaran@gmail.com Switch account

Not shared

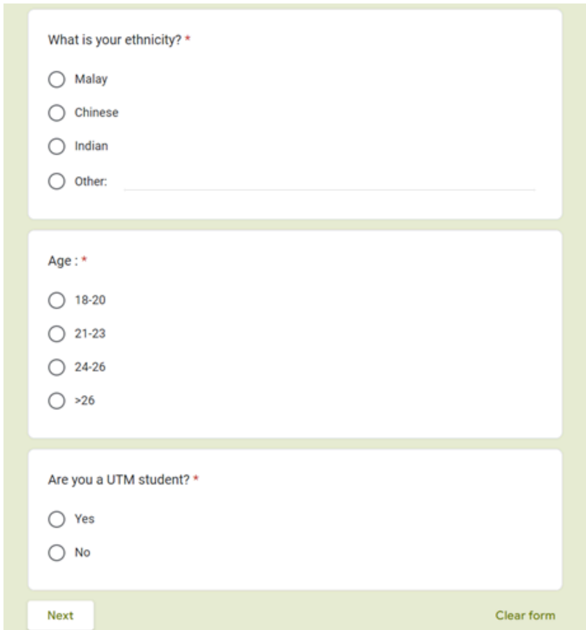
* Indicates required question

Email : *

Your answer

Gender : *

Male



What is your ethnicity? *

Malay

Chinese

Indian

Other:

Age : *

18-20

21-23

24-26

>26

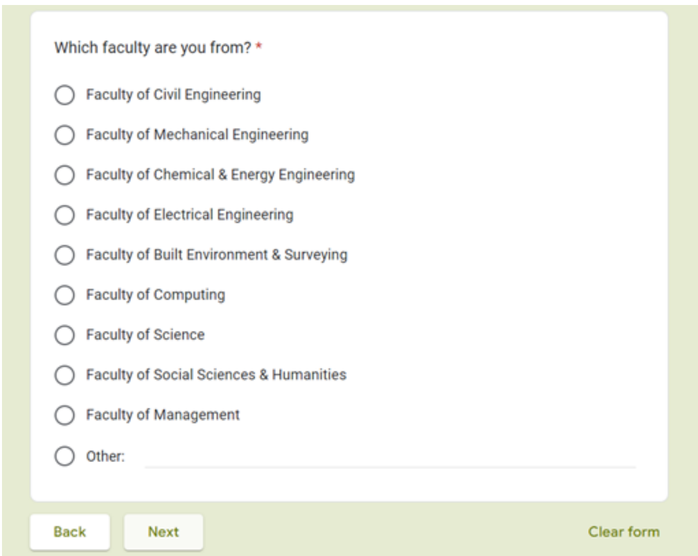
Are you a UTM student? *

Yes

No

Next

Clear form



Which faculty are you from? *

Faculty of Civil Engineering

Faculty of Mechanical Engineering

Faculty of Chemical & Energy Engineering

Faculty of Electrical Engineering

Faculty of Built Environment & Surveying

Faculty of Computing

Faculty of Science

Faculty of Social Sciences & Humanities

Faculty of Management

Other:

Back

Next

Clear form

Which university/college are you from? *

Your answer

Which faculty are you from? *

Your answer

Back Next Clear form

Where are you studying? *

- ◯ Perlis
- ◯ Penang
- ◯ Kedah
- ◯ Perak
- ◯ Selangor
- ◯ Melaka
- ◯ Negeri Sembilan
- ◯ Johor
- ◯ Pahang
- ◯ Terengganu
- ◯ Kelantan
- ◯ Sabah
- ◯ Sarawak
- ◯ Kuala Lumpur

Where is your hometown? *

- ◯ Perlis
- ◯ Penang

What is your primary transportation when traveling back home? *

- ◯ Train
- ◯ Bus
- ◯ Car
- ◯ Flight
- ◯ Other:

Which transportation did you use when you went back to your hometown from your study place ? *

- ◯ Train
- ◯ Bus
- ◯ Car
- ◯ Flight
- ◯ Other:

How much money did you allocate to spend to travel back to your hometown? *

- ◯ <RM 50
- ◯ RM 50-100

Was there any changes in your expenditure when purchasing the tickets than the usual? *

- ◯ Yes
- ◯ No

On a scale of 1 to 5, how did the cost of travelling affect your mode of transport? *

	1	2	3	4	5	
Least likely	◯	◯	◯	◯	◯	Most likely

How many hours did you spend on your way back to your hometown ? *

- ◯ < 1h
- ◯ 1h-3h
- ◯ 3h-5h
- ◯ 5h-7h
- ◯ 7h-10h
- ◯ > 10h

On a scale of 1 to 5, how comfortable were you with your mode of transport? *

	1	2	3	4	5	
Least likely	◯	◯	◯	◯	◯	Most likely

Do you prefer to air travel when you are going back to your hometown? *

- ◯ Yes
- ◯ No

Are there any factors that influence your choice of transportation such as convenience, comfort, environmental concerns? *

- ◯ Yes
- ◯ No

If yes, please state which factor it is. *

Your answer

Back Submit Clear form

It is known that every student looks forward to returning to their hometown during festive seasons and we wanted to know more about how festive seasons such as Eid affects the overall experience of travelling back home for students. Therefore, the purpose of our survey is to study the mode of transportation primarily used by students to return to their hometown every time which includes semester breaks and how certain circumstances affect the mode of students' transportation. We were expecting to know more deeply about whether festive seasons cause changes in travelling time, the cost of tickets and any other factors such as convenience and comfort and if so, what was the mode of transport chosen by students due to the changes.

DATA COLLECTION

1. The first data that were collected from the respondents were the general question.
   - The purpose of these basic demographic inquiries is to learn more about the various traits of respondents who intend to travel throughout the Eid vacation. It is important to comprehend demographics like age, gender, place of study, faculty or major, ethnicity, and gender in order to customise transportation services to the unique requirements and preferences of passengers. Your feedback will assist us in creating transport options that are inclusive and flexible enough to meet the needs of a wide range of travellers over the holiday season.

   a) Gender - categorical data
   b) Ethnicity - categorical data
   c) Average of ages - numerical data
   d) A UTM student - categorical data
   e) Faculty - categorical data
   f) University - categorical data
   g) Place of Study - categorical data

2. Question: Where is your hometown?
   - The purpose of the question above is to collect crucial information regarding the participants' places of origin. By proposing the following question, we hope to learn more about the respondents' geographic dispersion and pinpoint key centres from where people are travelling for the Eid vacation. This information is useful for arranging travel routes, calculating travel times, and making sure that passengers have access to enough facilities and services during this joyous time.

3. Question: What is your primary transportation when travelling back home?
   - The purpose of this question is to learn more about the main modes of transportation that respondents choose to use when travelling over the Eid holiday. Planning and optimising transportation services during peak travel hours requires an understanding of people's principal modes of transportation. Please choose the option that most closely matches the main means of transport for the Eid vacation, taking into consideration factors like cost, convenience, and personal preference. We will be able to better accommodate the most common means of transportation and improve the overall travel experience with your help in adjusting transportation services.

4. Question: Which transportation did you use when you went back to your hometown from your study place?
   - The purpose of this question is to collect particular information regarding the means of transportation that respondents used to return to their hometown for the Eid break from their study location. Gaining insight into the modes of transportation that people choose to use on this trip will help you better understand their travel habits and preferences. Kindly choose the option that matches the main mode of transportation utilised for returning to the hometown. In order to improve the overall travel experience during next Eid breaks, the response will aid in the analysis of travel behaviour and the optimisation of transport services.

5. Question: How much money did you allocate to spend to travel back to your hometown?
    - The purpose of this question is to learn more about the respondents' financial planning and budgeting for their travel expenses throughout the Eid vacation. It serves to show how much money was set aside or budgeted for the cost of returning home after the study place. This covers costs for things like travel fees, accommodation (if needed), meals, and any additional related expenses. The answer will be useful in determining whether a trip is affordable and in locating possible sources of funding or support. The question included a range of financial amounts.

6. Question: Was there any changes in your expenditure when purchasing the tickets than the usual?
    - The purpose of this question is to find out if respondents' typical spending patterns changed in any way when they bought tickets for their travel over the Eid holiday. It is intended to show whether the respondents saw any variations in the amount they paid for tickets in relation to what they usually spent on comparable travel at other periods. Changes might include increases, decreases, or no clear shift in spending. The answer will be useful in evaluating changes in ticket costs and comprehending how they affect travellers' spending plans.

7. Question: On a scale of 1 to 5, How did the cost of travelling affect your mode of transport?

    - This question seeks to gauge the impact of travel costs on respondents' choice of transportation mode. The respondents answered, on a scale of 1 to 5, how significantly the cost of travelling influenced their decision regarding which mode of transport to use for their Eid break travel.
    - 1: The cost had minimal to no influence on the choice of transportation.
    - 2: The cost had a slight influence on the choice, but it was not a determining factor.
    - 3: The cost moderately influenced the choice of transportation mode.
    - 4: The cost significantly influenced the choice, playing a major role in determining the mode of transport.
    - 5: The cost had a decisive impact on the choice of transportation, being the primary factor in selecting the mode of travel.
    - The selection of the rating that best reflects how the cost of travelling influenced their decision-making process. The response will help in understanding the importance of cost considerations in travellers' mode of transport selection during the Eid break.

8. Question: How many hours did you spend on your way back to your hometown?

    - The purpose of this question is to learn more about how long respondents' trips took to return to their hometowns for the Eid holiday. This shows the overall amount of time travelled, including any necessary layovers or stops, from their departure point (study place) to their destination (hometown). It will be easier to comprehend the time commitments associated with travelling during the Eid holiday and to optimise transport services if a precise estimate of journey time is provided.

9. Question: On a scale 1 to 5, How comfortable were you with your mode of transport?

    - This question aims to assess respondents' comfort levels with the mode of transportation they used for their journey back to their hometown during the Eid break. The respondents answered, on a scale of 1 to 5, how comfortable they felt during their travel experience.

- 1: Very uncomfortable
- 2: Uncomfortable
- 3: Neither comfortable nor uncomfortable
- 4: Comfortable
- 5: Very comfortable

- The selection of the rating that best reflects their level of comfort with the mode of transport they utilised. Factors influencing comfort may include seating arrangements, cleanliness, amenities, safety measures, and overall travel experience. The response will help in understanding the comfort preferences of travellers and improving transportation services to enhance passenger satisfaction.

10. Question: Do you prefer air travel when you are going back to your hometown?

- The purpose of this question is to learn more about the respondents' preferred methods of air travel for their Eid break return to their hometown. The question reveals if they would rather go by air for this particular trip. The answer will be helpful in determining the demand for air transportation services as well as offering insightful information about how popular air travel is with vacationers over the Eid break. Regardless of their inclination towards air travel, their input is crucial in customising modes of transportation to better suit their requirements and tastes.

11. Question: Are there any factors that influence your choice of transportation such as convenience, comfort, environmental concerns?

- The purpose of the question is to know whether the respondents have any problem or factors that affect their journey to go back to their hometown.

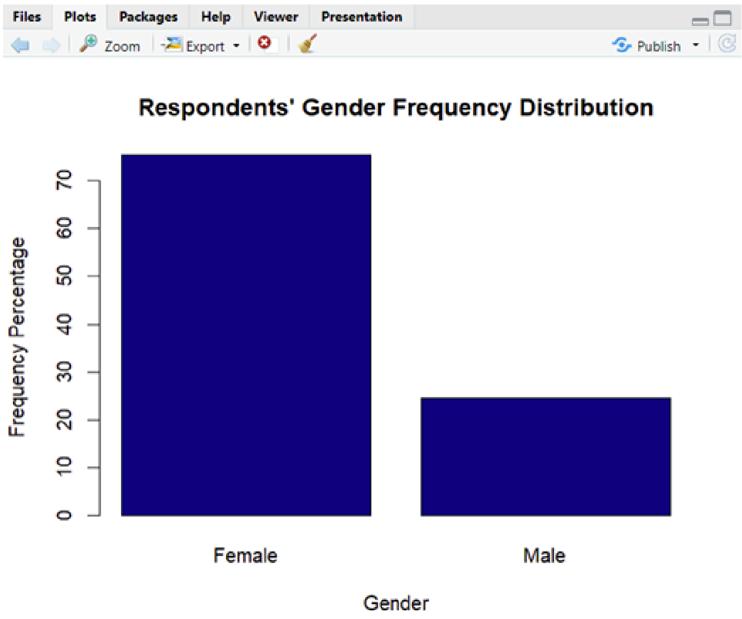
12. Question: If Yes, please state which factor it is

- To indicate that there are factors influencing the choice of transportation in the previous question, we would like to know which specific factor or factors are most important to the respondents. It is stated whether convenience, comfort, environmental concerns, or any combination of these factors influence their transportation choice. The response helps understand their priorities better and tailor transportation options accordingly.

DATA ANALYSIS

A) CATEGORICAL DATA:

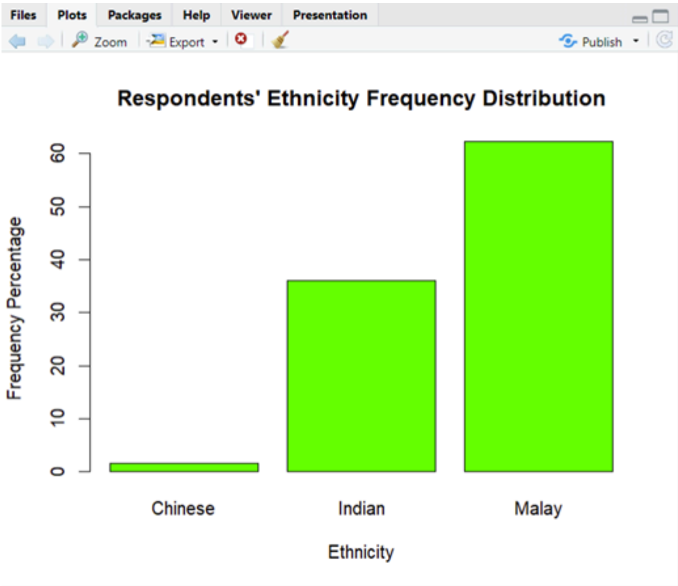
- **GRAPHS**



	Gender	Freq
1	Female	75.40984
2	Male	24.59016

Figure 1.1

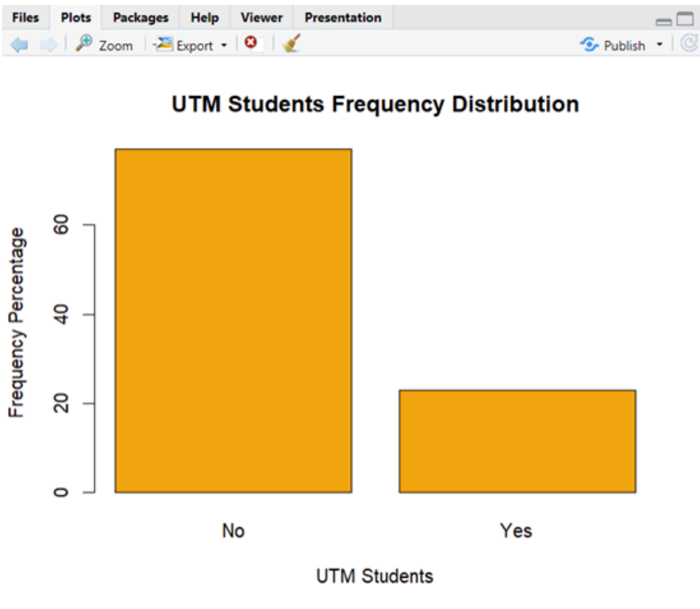
There were a total of 15 male and 46 female respondents that answered our questionnaire. Figure 1.1 shows the percentage distribution of 24.59% and 75.41% for male and female respondents respectively.



	Ethnicity	Freq
1	Chinese	1.639344
2	Indian	36.065574
3	Malay	62.295082

Figure 1.2

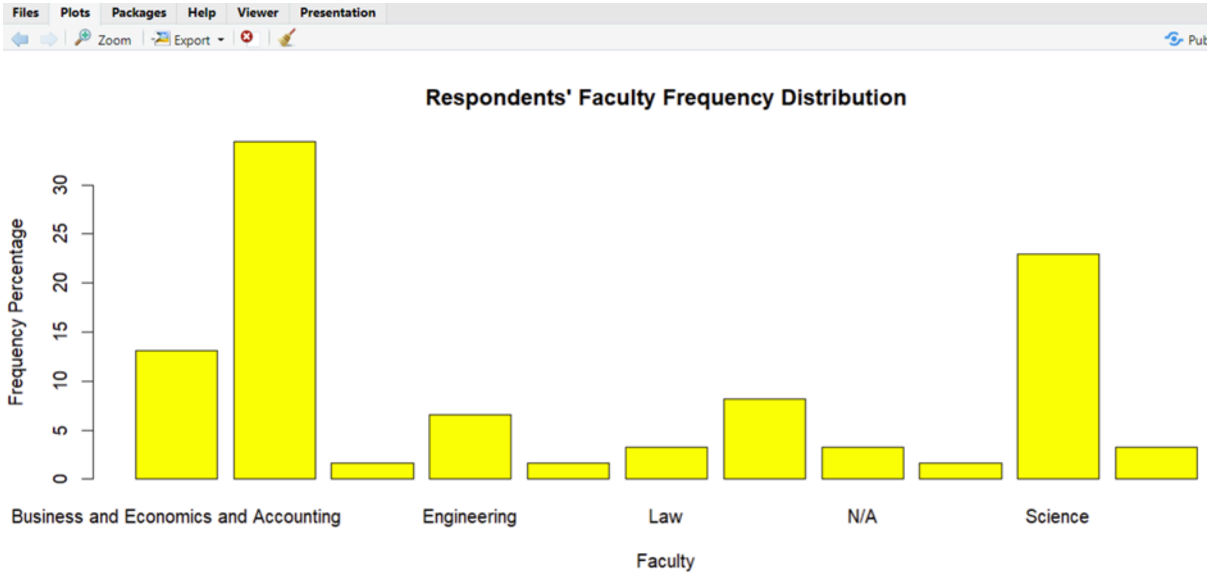
There were a total of 1 Chinese respondent, 22 Indian respondents and 38 Malay respondents that answered our questionnaire. Figure 1.2 shows the percentage distribution of 1.64% Chinese respondents, 36.06% Indian respondents and 62.30% Malay respondents that answered our questionnaire.



	UTM.student	Freq
1	No	77.04918
2	Yes	22.95082

Figure 1.3

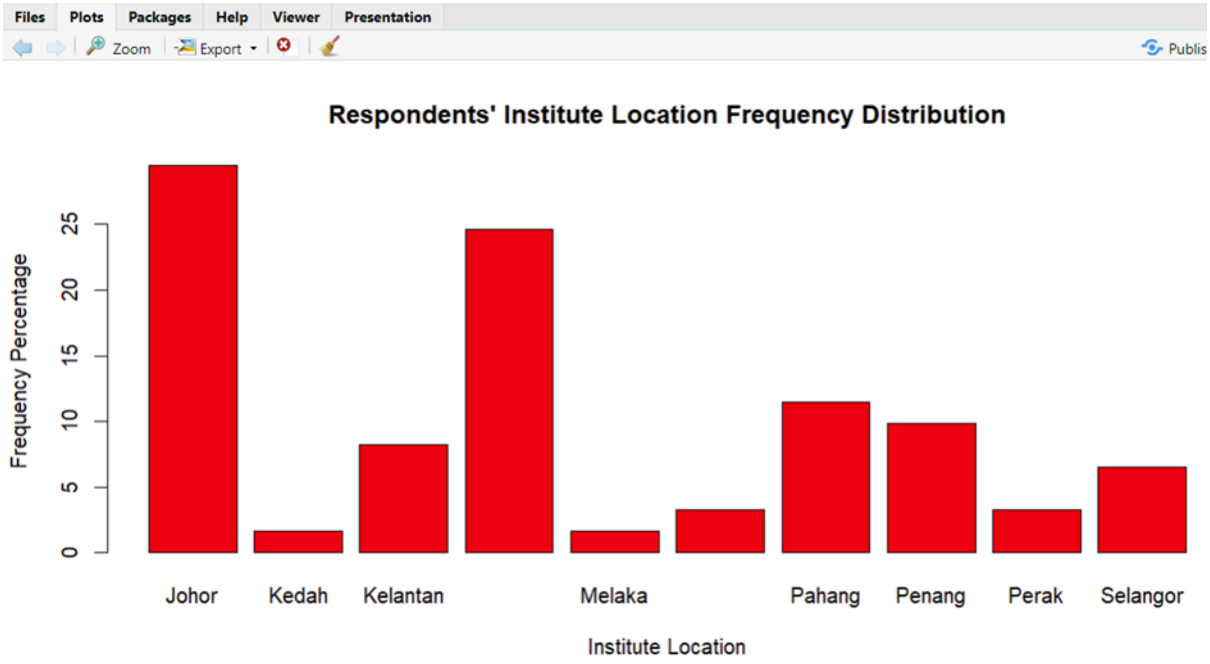
Out of 61 respondents, 14 students were from UTM and 47 students were non-UTM students. Figure 1.3 shows the frequency distribution of the respondents where 77.05% students are non-UTM students whereas 22.95% students are UTM students. This shows that our survey has been well dispersed to students from all over Malaysia to give in their responses.



	Faculty	Freq
1	Business and Economics and Accounting	13.114754
2	Computing	34.426230
3	Diploma	1.639344
4	Engineering	6.557377
5	Language Academy	1.639344
6	Law	3.278689
7	Medicine, Dentisry and Nursing	8.196721
8	N/A	3.278689
9	Pengajian Islam dan Sains Sosial	1.639344
10	Science	22.950820
11	Social Sciences & Humanities	3.278689

Figure 1.4

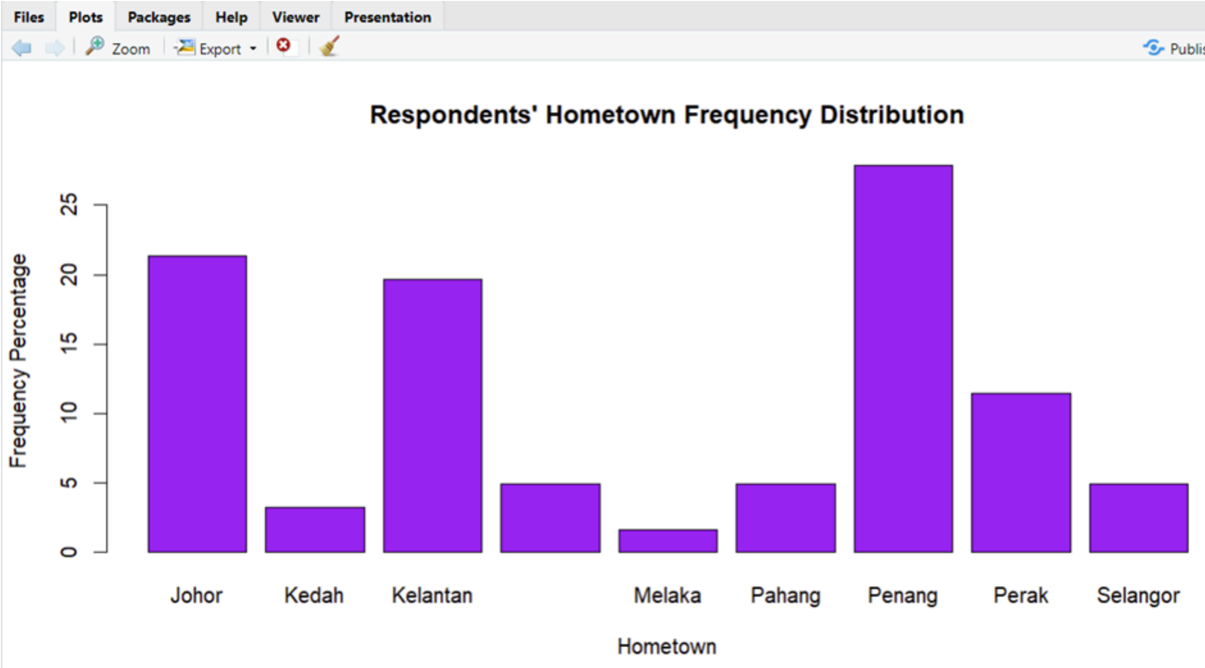
There were a total of 11 different faculties that our respondents who answered our questionnaire were part of. 8 students from the Faculty of Business and Economics and Accounting, 21 students from the Faculty of Computing, 4 students from the Faculty of Engineering, 5 students from the Faculty of Medicine, Dentistry and Nursing and 14 students from the Faculty of Science. There are 2 students each from Faculty of Social Science and Humanities, Faculty of Law and also N/A as they're currently working. We also have 1 student each from the Faculty of Islamic Studies, Language Academy and Diploma Studies. Figure 1.4 shows the frequency distribution of 13.11% for Business and Economics and Accounting students, 34.43% for Computing students, 6.56% for Engineering students, 8.19% for Medicine, Dentistry and Nursing students and 22.95% for Science students. There are 3.28% of students each in Faculty of Law, Faculty of Social Science and Humanities students and N/A and finally, 1.64% students each in Diploma Studies, Language Academy and Faculty of Islamic Studies. This bar chart shows that we've gathered a sample of students from almost all faculties present in all universities.



	Institute.Location	Freq
1	Johor	29.508197
2	Kedah	1.639344
3	Kelantan	8.196721
4	Kuala Lumpur	24.590164
5	Melaka	1.639344
6	Negeri Sembilan	3.278689
7	Pahang	11.475410
8	Penang	9.836066
9	Perak	3.278689
10	Selangor	6.557377

Figure 1.5

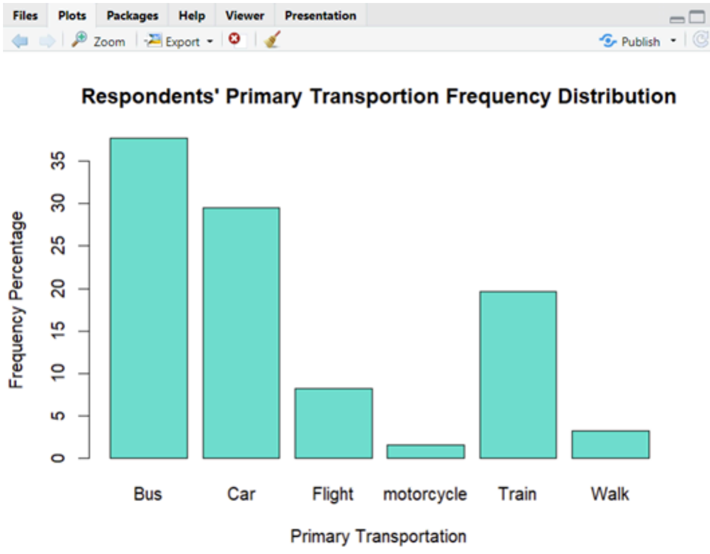
Out of 61 students, 18 students are pursuing their studies in Johor, 5 students in Kelantan, 15 students in Kuala Lumpur, 7 students in Pahang, 6 students in Penang and 4 students in Selangor. There are 2 students each pursuing their studies in Perak and Negeri Sembilan and 1 student each in Kedah and Melaka. 0 students are pursuing their studies in Perlis, Terengganu, Sabah and Sarawak. Therefore, those four states are not included in the bar chart. Figure 1.5 shows the frequency distribution of respondents' institute location. 29.51% of students are studying in Johor, 8.19% of students in Kedah, 24.59% of students in Kuala Lumpur, 11.47% students in Pahang, 9.84% of students in Penang and 6.56% students in Selangor. There are 3.28% of students each in Perak and Negeri Sembilan and finally, 1.64% students each in Kedah and Melaka. There are 0% students in Perlis, Terengganu, Sabah and Sarawak. Although we failed to gather data from students who are pursuing their studies in Perlis, Terengganu, Sabah and Sarawak, our survey still has covered students who are studying all across Malaysia.



	Hometown	Freq
1	Johor	21.311475
2	Kedah	3.278689
3	Kelantan	19.672131
4	Kuala Lumpur	4.918033
5	Melaka	1.639344
6	Pahang	4.918033
7	Penang	27.868852
8	Perak	11.475410
9	Selangor	4.918033

Figure 1.6

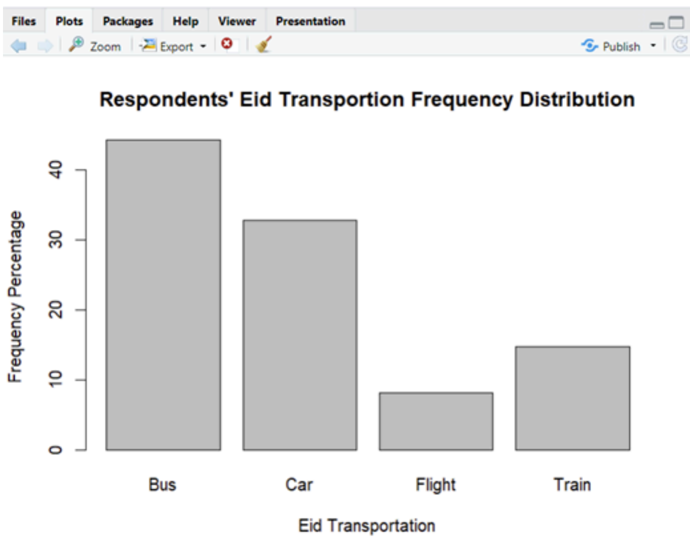
Out of 61 students, 13 students reside in Johor, 2 students in Kedah, 12 students in Kelantan, 1 student in Melaka, 17 students in Penang and 7 students in Perak. There are 3 students each residing in Kuala Lumpur, Pahang and Selangor. 0 students are residing in Perlis, Negeri Sembilan, Terengganu, Sabah and Sarawak. Therefore, those five states are not included in the bar chart. Figure 1.6 shows the frequency distribution of respondents' hometown. 21.31% of students are staying in Johor, 3.28% of students in Kedah, 19.67% of students in Kelantan, 1.64% students in Melaka, 27.87% of students in Penang and 11.47% students in Perak. There are 4.92% of students each in Kuala Lumpur, Pahang and Selangor. There are 0% students in Perlis, Negeri Sembilan, Terengganu, Sabah and Sarawak. Although we failed to gather data from students who are staying in Perlis, Negeri Sembilan, Terengganu, Sabah and Sarawak, we gathered data from students who travel quite often to their institute location which is the main purpose of our survey.



	Primary.Transportation	Freq
1	Bus	37.704918
2	Car	29.508197
3	Flight	8.196721
4	motorcycle	1.639344
5	Train	19.672131
6	Walk	3.278689

Figure 1.7

There are 23 students who primarily travel by bus, 18 students who travel by car, 5 students who travel by flight, 1 student who travel by motorcycle, 12 students who travel by train and 2 students who walk back to their home. Figure 1.7 shows the percentage frequency whereby 37.70% students choose bus as their primary transportation, 29.51% students who travel by car, 8.20% students who travel by flight, 1.64% students who travel by motorcycle, 19.67% students who travel by train and finally 3.28% students who walk back home.

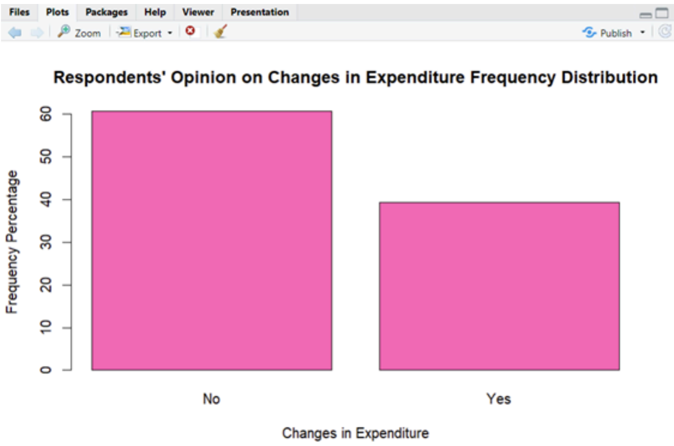


	Eid.Transportation	Freq
1	Bus	44.262295
2	Car	32.786885
3	Flight	8.196721
4	Train	14.754098

Figure 1.8

For the past Eid break, 27 students travelled home by bus, 20 students by car, 5 students by flight and 9 students by train. Figure 1.8 shows the percentage distribution whereby 44.26% students travelled home during the Eid break by bus, 32.79% students travelled by car, 8.20% students by flight and 14.75% students by train. It is evident that the mode of transport has changed for some of the students.

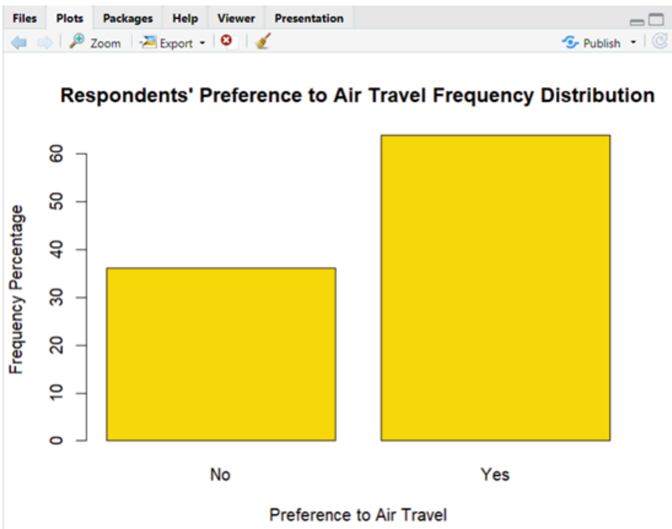
For example, students who primarily opted to travel back home by walking or motorcycle, opted for different mode of transportation for this Eid break. The percentage of students travelling by bus and car has increased compared to the percentage in primary transportation by 6.56% for bus and 3.28% for car. The percentage of students travelling back home by flight remains constant which is 8.20%. lastly, the percentage of students travelling by train has decreased by 4.92%.



	Changes.in.Expenditure	Freq
1	No	60.65574
2	Yes	39.34426

Figure 1.9

Out of 61 respondents, 37 students think there are no changes in expenditure when travelling back home during festive seasons whereas 24 students think there are changes in expenditure. Figure 1.9 shows the percentage distribution of the respondent's opinion on changes in expenditure where 60.66% students said no and 39.34% students said yes. Students who have said no might be travelling by train or bus where the cost of ticket rarely fluctuates whereas students who said yes might be travelling by car and flight as for flight, the date of the ticket influences the price and as for car, the toll fees and petrol fees can change too.



	Preference.to.air.travel	Freq
1	No	36.06557
2	Yes	63.93443

Figure 1.10

If given the chance to air travel, 39 students preferred to air travel whereas 22 students would turn down the offer. Figure 1.10 shows the percentage distribution of the preference of respondents' to air travel whereby 36.07% students would not travel by flight whereas 63.93% would travel by flight. This proves that the cost of the ticket is a major factor contributing to the mode of students' transport to travel back home.

- **NUMERICAL**

Respondents' age

AGE	FREQUENCY
18-20	**45**
21-23	**7**
24-26	**1**
>26	**8**
TOTAL	**61**

18-20: This age group comprises the majority of respondents, with 45 individuals falling within this range. It suggests that the survey primarily targeted UTM students.

21-23: Seven respondents fall within this age range. While smaller in number compared to the 18-20 group, it still represents a notable portion of the survey population.

24-26: Only one respondent falls within this age range, indicating a much smaller representation in the survey.

>26: Eight respondents are older than 26 years old. While still a relatively small portion of the total respondents, it's noteworthy that there are respondents from a broader age range, indicating some diversity in the sample since we also seek for respondents' outside of UTM.

Amount of money the respondents' allocated to spend on transportation

MONEY SPENT ON TRANSPORTATION	FREQUENCY
< RM 50	**25**
RM50 – RM100	**23**

RM100 – RM200	**8**
>RM200	**5**

< RM 50: The majority of respondents, 25 individuals, allocated less than RM 50 for transportation expenses. This suggests that a significant portion of the respondents prioritized affordability when planning their travel expenses.

RM 50 – RM 100: 23 respondents allocated between RM 50 and RM 100 for transportation. This range represents a slightly higher budget for transportation, indicating a willingness to spend a bit more for added convenience or comfort.

RM 100 – RM 200: Eight respondents allocated between RM 100 and RM 200 for transportation expenses. This indicates a willingness to invest more in transportation, likely for longer journeys or for higher comfort levels.

> RM 200: Five respondents allocated more than RM 200 for transportation. This suggests that a smaller portion of respondents had higher budgets for transportation, possibly for flights or premium transportation services.

Overall, the data reflects a range of budget allocations among respondents, with the majority opting for lower-cost options, but some willing to spend more for added convenience, comfort, or faster travel options.

Effects of cost travelling on respondents' mode of transportation

LIKERT SCALE	FREQUENCY	PROPORTION	PERCENTAGE
1	8	8/61	13.11%
2	5	5/61	8.20%
3	26	26/61	42.62%
4	13	13/61	21.31%
5	9	9/61	14.75%

8 out of 61 respondents, accounting for approximately 13.11%, rated the effect of cost on their mode of transportation as 1. This indicates that a small proportion of respondents felt that cost had a minimal or negligible impact on their choice of transportation.

5 out of 61 respondents, representing approximately 8.20%, rated the effect of cost as 2. This suggests that a slightly smaller proportion of respondents perceived a low impact of cost on their choice of transportation.

26 out of 61 respondents, making up approximately 42.62%, rated the effect of cost as 3. This indicates that a significant portion of respondents perceived a moderate impact of cost on their choice of transportation.

13 out of 61 respondents, representing approximately 21.31%, rated the effect of cost as 4. This suggests that a substantial portion of respondents perceived a high impact of cost on their choice of transportation.

9 out of 61 respondents, also accounting for approximately 14.75%, rated the effect of cost as 5. This indicates that a similar proportion of respondents perceived a very high impact of cost on their choice of transportation as those who rated it as 4.

Overall, the data suggests that while some respondents felt that cost had minimal impact on their choice of transportation, the majority perceived varying degrees of influence, with a significant portion acknowledging a moderate to high impact of cost on their mode of transportation selection.

Hours spent on their way back to their hometown

TIME INTERVAL	FREQUENCY
>10 HOURS	**4**
7 HOURS – 10 HOURS	**9**
5 HOURS – 7 HOURS	**8**
3 HOURS – 5 HOURS	**19**
1 HOUR – 3 HOURS	**9**
< 1 HOUR	**12**

> 10 HOURS: Four respondents reported spending more than 10 hours traveling back to their hometowns. This suggests that for some individuals, longer travel times may be a result of choosing lower-cost transportation options.

7 HOURS – 10 HOURS: Nine respondents reported travel times between 7 and 10 hours. This range likely represents journeys that span a considerable distance but may still be influenced by cost considerations.

5 HOURS – 7 HOURS: Eight respondents reported travel times between 5 and 7 hours. This range may indicate moderately long journeys, possibly influenced by both cost and convenience factors.

3 HOURS – 5 HOURS: Nineteen respondents reported travel times between 3 and 5 hours. This range likely represents a balance between cost-effective transportation and reasonable travel time.

1 HOUR – 3 HOURS: Nine respondents reported travel times between 1 and 3 hours. This range suggests relatively shorter travel distances or possibly the use of faster transportation modes, which may come with higher costs.

< 1 HOUR: Twelve respondents reported travel times of less than 1 hour. This indicates very short travel distances, possibly within the same city or nearby areas, where cost considerations may not be as significant.

How comfortable are the respondents with their chosen modes of transportation

LIKERT SCALE	FREQUENCY	PROPORTION	PERCENTAGE
1	**2**	**2/61**	**3.28%**
2	**9**	**9/61**	**14.75%**
3	**19**	**19/61**	**31.15%**
4	**12**	**12/61**	**19.67%**
5	**19**	**19/61**	**31.15%**

2 out of 61 respondents, representing approximately 3.28%, rated their comfort level as 1. This suggests that a very small proportion of respondents were highly uncomfortable with their chosen modes of transportation.

9 out of 61 respondents, accounting for approximately 14.75%, rated their comfort level as 2. This indicates a slightly larger proportion of respondents who expressed some discomfort but not to the highest extent.

19 out of 61 respondents, making up approximately 31.15%, rated their comfort level as 3. This suggests that a significant portion of respondents felt neutral or moderately comfortable with their chosen modes of transportation.

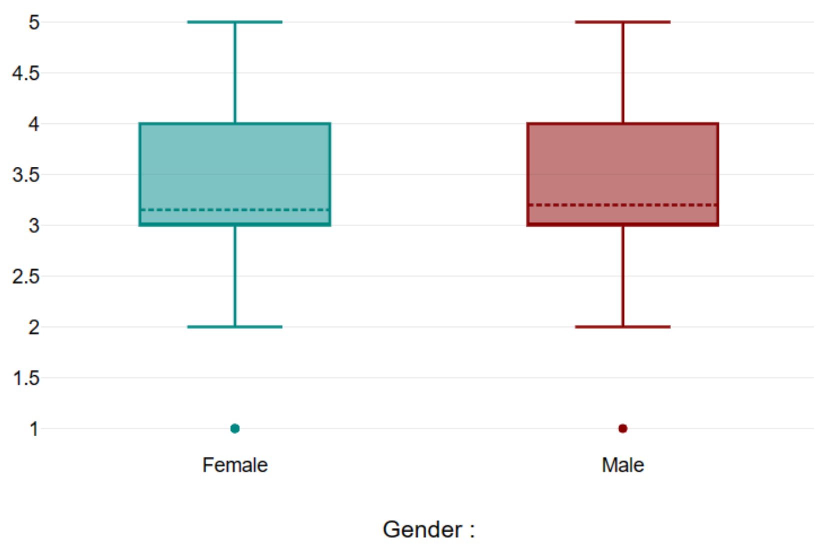
12 out of 61 respondents, representing approximately 19.67%, rated their comfort level as 4. This indicates a substantial portion of respondents who felt quite comfortable with their chosen modes of transportation.

19 out of 61 respondents, also representing approximately 31.15%, rated their comfort level as 5. This suggests that an equal proportion of respondents felt highly comfortable with their chosen modes of transportation as those who rated it as 3.

Overall, the data reveals that while a minority of respondents expressed discomfort, the majority felt either neutral, moderately comfortable, or highly comfortable with their chosen modes of transportation.

B) QUANTITATIVE DATA

- **GRAPHS**

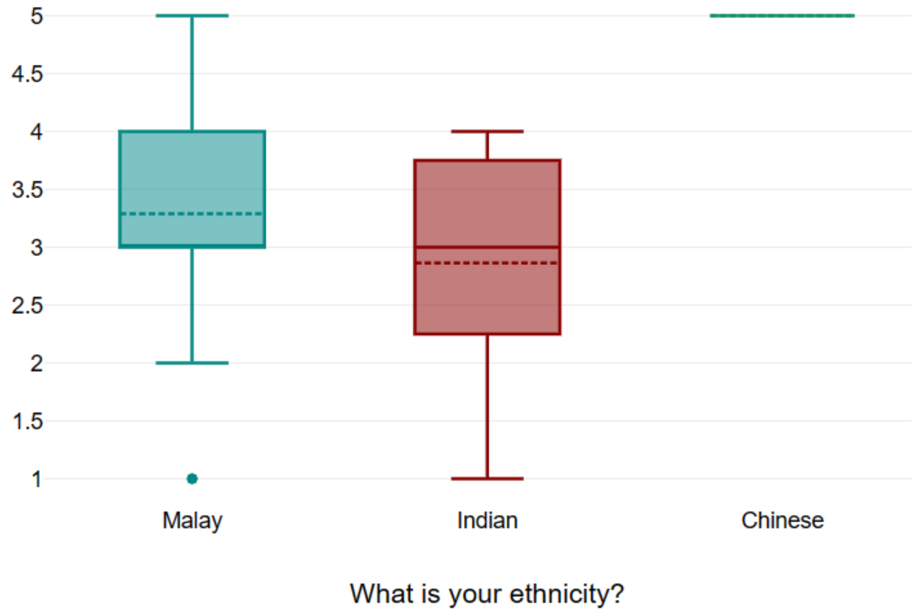


The measures of central tendency are:

Mean for males: approximately 24.59%

Mean for females: approximately 75.41%

Mode: "female"



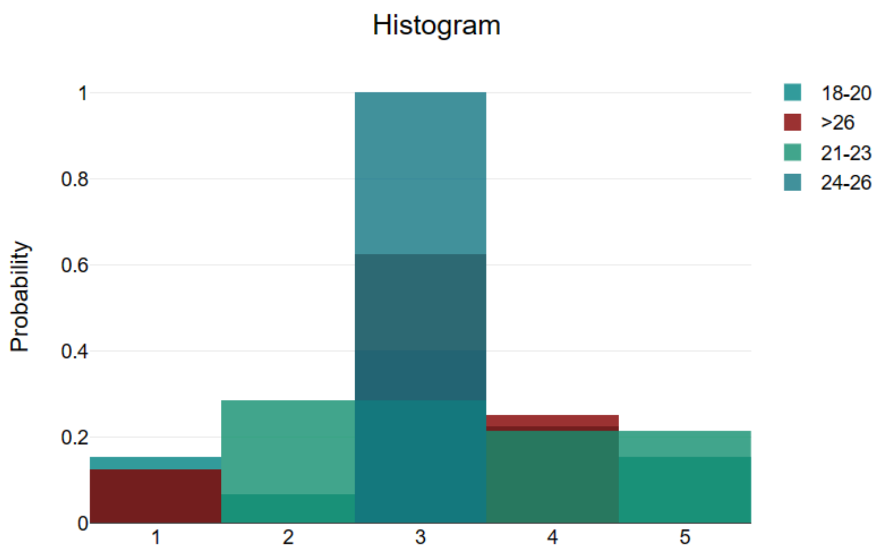
The measures of central tendency are:

Mean for Chinese: approximately 1.64%

Mean for Indian: approximately 36.07%

Mean for Malay: approximately 62.30%

Mode: "Malay"



AGE

The measures of central tendency are:

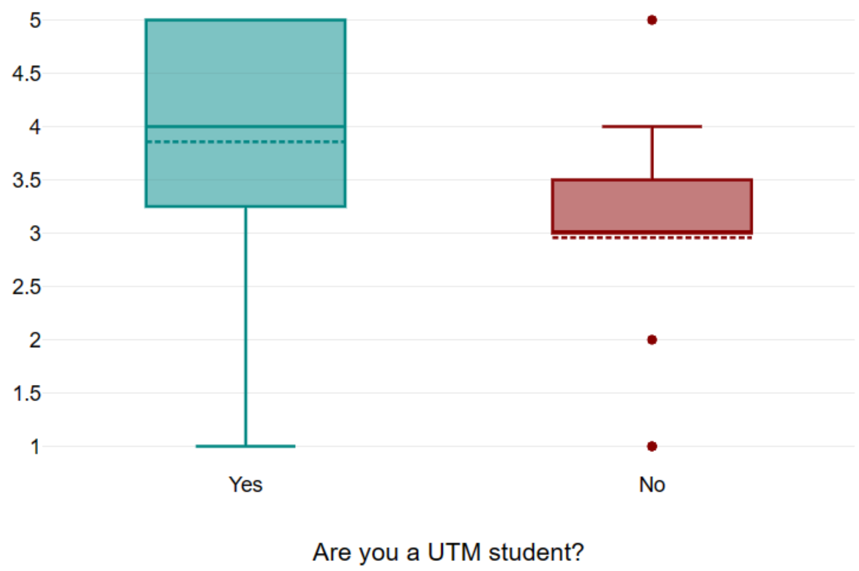
Mean for age group 18-20: approximately 73.77%

Mean for age group 21-23: approximately 11.48%

Mean for age group 24-26: approximately 1.64%

Mean for age group >26: approximately 13.11%

Mode: "18-20"

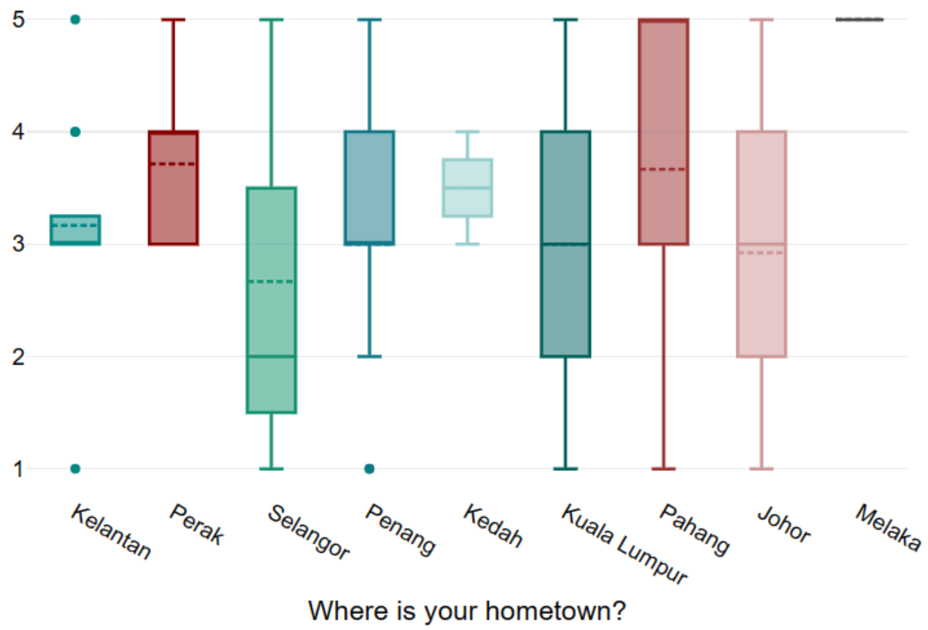


The measures of central tendency are:

Mean for UTM students: approximately 22.95%

Mean for non-UTM students: approximately 77.05%

Mode: "non-UTM students



The measures of central tendency are:

Mean for Johor: approximately 21.31%

Mean for Kedah: approximately 3.28%

Mean for Kelantan: approximately 19.67%

Mean for Melaka: approximately 1.64%

Mean for Penang: approximately 27.87%

Mean for Perak: approximately 11.48%

Mean for Kuala Lumpur, Pahang, and Selangor: approximately 4.92%

Mode: "Penang"

CONCLUSION

Based on the data analysis provided, it's evident that various factors influence individuals' choice of transportation when travelling back to their hometowns. These factors include convenience, comfort, cost, time, environmental concerns, and personal preferences.

The majority of respondents seem to prioritise comfort and convenience when selecting their mode of transportation. Many opt for buses due to their affordability and availability, while others prefer cars for their flexibility and comfort. Additionally, a significant number of respondents mention environmental concerns as a factor influencing their choice, indicating a growing awareness of sustainability issues.

Interestingly, despite the availability of other options, such as trains and flights, buses and cars remain the preferred choices for most respondents. This suggests that while factors like time-saving and comfort are essential, affordability and accessibility play significant roles in decision-making.

In conclusion, the data highlights the complexity of factors influencing individuals' transportation choices when travelling back to their hometowns. Understanding these factors can inform policymakers and transportation providers in catering to the diverse needs of travellers and promoting more sustainable and efficient transportation options.